```
import pandas as pd
import numpy as np

data = pd.read_csv("california_housing_train.csv")
data.head()
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	hous
0	-114.31	34.19	15.0	5612.0	1283.0	1015.0	
1	-114.47	34.40	19.0	7650.0	1901.0	1129.0	
2	-114.56	33.69	17.0	720.0	174.0	333.0	
3	-114.57	33.64	14.0	1501.0	337.0	515.0	
4	-114.57	33.57	20.0	1454.0	326.0	624.0	

Next steps: Generate code with data New interactive sheet

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17000 entries, 0 to 16999
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   longitude           17000 non-null  float64
 1   latitude            17000 non-null  float64
 2   housing_median_age  17000 non-null  float64
 3   total_rooms         17000 non-null  float64
 4   total_bedrooms      17000 non-null  float64
 5   population          17000 non-null  float64
 6   households          17000 non-null  float64
 7   median_income       17000 non-null  float64
 8   median_house_value  17000 non-null  float64
dtypes: float64(9)
memory usage: 1.2 MB
```

```
data.shape
```

```
(17000, 9)
```

```
data.isnull().sum()
```

	0
longitude	0
latitude	0
housing_median_age	0
total_rooms	0
total_bedrooms	0
population	0
households	0
median_income	0
median_house_value	0

dtype: int64

```
x = data.drop("median_house_value", axis=1)
y = data["median_house_value"]

#train-test split

from sklearn.model_selection import train_test_split

x_train,x_test,y_train,y_test = train_test_split(
    x,
    y,
    test_size = 0.2,
    random_state = 42

)

#model training
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(x_train,y_train)
print("Model Trained Successfully")
```

```
Model Trained Successfully
```

```
#Prediction

predictions = model.predict(x_test)
print("predictions:",predictions[:10])
```

```
predictions: [143770.39502963 398615.57056493  86341.10306742 148534.35353345
 147202.29808602 279272.30068644 154882.64433797 139598.86735209
 352948.93013951 259959.28700011]
```

```
#Evaluation
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score

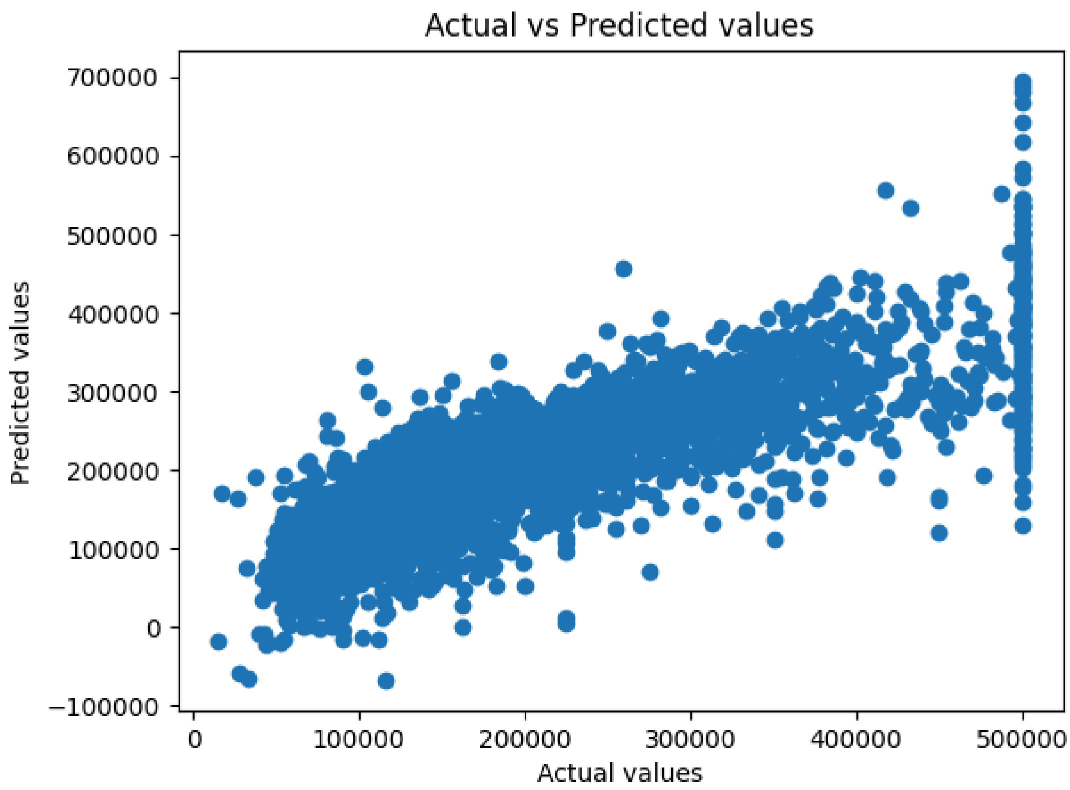
mae = mean_absolute_error(y_test,predictions)
mse = mean_squared_error(y_test,predictions)
```

```
r2 = r2_score(y_test,predictions)

print("MAE:",mae)
print("MSE:",mse)
print("R2 score:",r2)
```

```
MAE: 49983.47465122931
MSE: 4634658406.223264
R2 score: 0.6636396350243869
```

```
import matplotlib.pyplot as plt
plt.scatter(y_test,predictions)
plt.xlabel("Actual values")
plt.ylabel("Predicted values")
plt.title("Actual vs Predicted values")
plt.show()
```



Close

```
**What is regression?**
It measures relationship between a dependent
variable and one or more independent variables

**Difference between regression and
classification?**
classification predicts discrete value or
discrete output. Regression predict continuous
values or continuous output.

**What is dependent variable?**
A dependent variable is the output variable
that we want to predict using machine learning

**What is overfitting?**
In machine learning overfitting occurs when a
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```
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**Explain regression line.**
A regression line is defined as a straight
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a dependent variable(Y) and an independent
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```

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